1. You have a coupler that divides an input signal equally among 16 outputs. It has no excess loss. If the input signal is —10 dBm, what is the output at any one port?
a. —12 dBm
b. —20dBm
c. —22dBm
d. —26dBm
e. —30 dBm

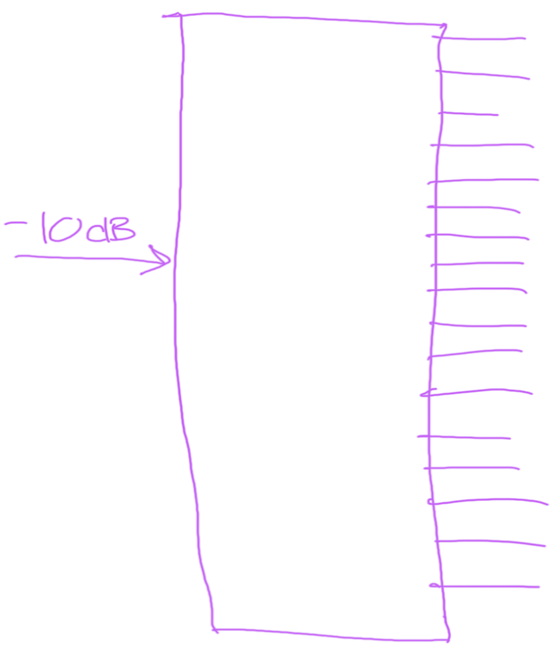
8:07

Notes

HWK 14

(1) 16 outputs
input -10dB
no excess loss

output of any port?



(A) dBm → dB → W

8:08

Notes

(A) dBm → dB → W

$dB = 10 \log_{10}(P)$

$\frac{-10dB}{10} = \frac{\cancel{10} \log_{10}(P)}{\cancel{10}}$

$-1dB = \log_{10}(P)$

$10^{-1} = \cancel{10^{(\log_{10})}}(P)$

$10^{-1} = \text{Power} = 0.1W$

(B) $\frac{0.1W}{16} = 6.25mW$

(C) $dBm = 10\log_{10}(6.25 \times 10^{-3} W)$

$= \boxed{-22.04 dBm}$

2. A 1 X 20 coupler has output signals of —30 dBm at every port if the input signal is —10 dBm. What is its excess loss?

a. 0 dB

b. 1dB

c. 2 dB

d. 4.2 dB

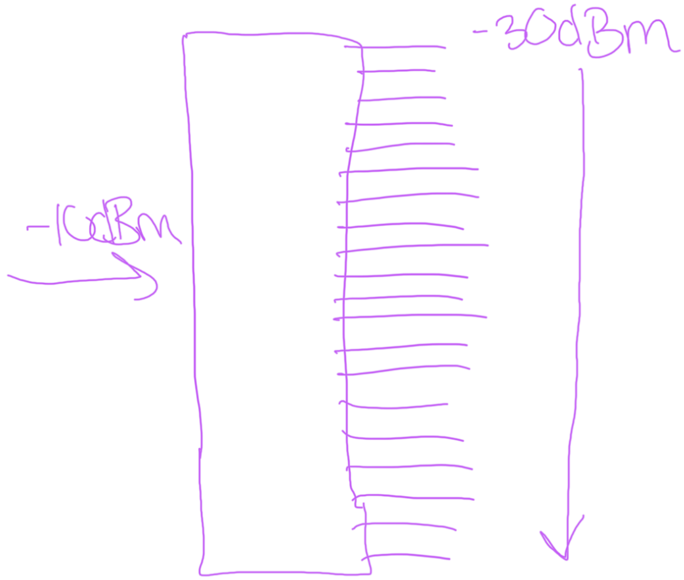
e. 7 dB

2) 1X20 OC

-30dBm = out

-10dBm = in

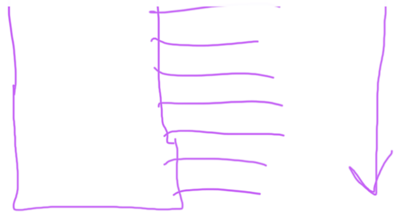
excess loss?



A) $EL = -10\log\left(\frac{\text{Total output}}{\text{input}}\right)$

↳ $10^{\left(\frac{dBm}{10}\right)} \to 10^{\left(\frac{-10}{10}\right)}$

↳ $10^{-1} = 0.1W$



A) $EL = -10\log\left(\frac{\text{Total output}}{\text{input}}\right)$

↳ $10^{\left(\frac{dBm}{10}\right)} \rightarrow 10^{\left(\frac{-10}{10}\right)}$

↳ $10^{-1} = 0.1W$

B) $10^{\left(\frac{-30}{10}\right)} \rightarrow 0.001W$

or $1mW$

C) $1mW \times 20 = 20mW$

D) $-10\log\left(\frac{20\cdot10^{-3}W}{0.1W}\right)$

$= 6.9897 \approx \boxed{7dB}$

3. A coupler splits an input signal between two ports with a 90/10 ratio. If the input signal is —20 dBm and the coupler has no excess loss, what is the output at the port receiving the smaller signal?

a. —21 dBm

b. —29 dBm

c. —30 dBm

d. —31 dBm

e. —110 dBm

3 splits btwn 2 ports

90/10

in = -20dBm

no EL

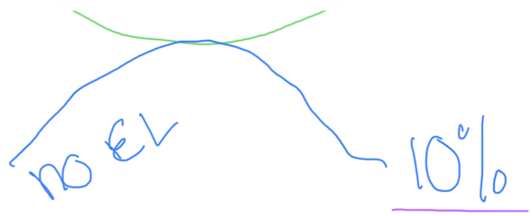
output @ port receiving smaller signal?

A input → output

↳ 10% of output

↳ $10^{\left(\frac{-20}{10}\right)} = 0.01\,W$

↳ 10% → 0.1



A input → output

↳ 10% of output

↳ $10^{\left(\frac{-20}{10}\right)} = 0.01\,W$

↳ 10% → 0.1

↳ $0.1 \times 0.01 = 0.001\,W$

B $10\log(0.001) = \boxed{-30\,dBm}$

4. What type of coupler could distribute identical signals to 20 different terminals?
a. T coupler
b. tree coupler
c. star coupler
d. M X N coupler
e. wavelength-selective coupler

5 . What type of coupler divides one input signal between two output channels?
a. T coupler
b. tree coupler
c. star coupler
d. M X N coupler
e. wavelength-selective coupler

6. You find a coupler with four ports and no label on it. You measure attenuation from port 1 to the other three ports. The values are:—40 dB to port 2, 3 dB to port 3, 3 dB to port 4. What type of coupler do you have?
a. star coupler with three unequal outputs
b. tree coupler with three unequal outputs
c. a directional 2-by-2 coupler with two inputs and two outputs
d. a nondirectional T coupler
e. a broken coupler

7 . A Y coupler that equally divides light between two outputs has a 3-dB loss on each channel. What is the right explanation?
a. The 3-dB figure is excess loss.
b. Half the photons that enter the coupler go out each output, corresponding to a 3-dB loss on each channel.
c. The coupler polarizes the light going out each port, causing 3-dB loss.
d. Every coupler has at least 3-dB loss no matter how it divides the input light.
e. The optics are dirty, causing loss of half the input light.

8. Evanescent waves cause light energy to transfer between channels in what type of coupler?
a. planar-waveguide coupler
b. single-mode fused-fiber coupler
c. bulk optical coupler
d. multimode fused-fiber coupler
e. a and b
f. b and d

9. An attenuator
a. is a filter that blocks one wavelength but transmits others.
b. polarizes input light, causing loss of the other polarization.

c. reduces light intensity evenly across a range of wavelengths.
d. selectively blocks photons produced by spontaneous emission.